Chapter 15, Problem 15PP

Problem

Solve the following differential equation using the Laplace transform method.

$$\frac{d^2v(t)}{dt^2} + 4\frac{dv(t)}{dt} + 4v(t) = 7e^{-t}$$

Step-by-step solution

Step 1 of 6

The differential equation is,

$$\frac{d^2v(t)}{dt^2}+4\frac{dv(t)}{dt}+4v(t)=7e^{-t}$$

Apply the Laplace transform of each term in the differential equation.

$$L\left\{\frac{d^2v(t)}{dt^2}\right\}+4L\left\{\frac{dv(t)}{dt}\right\}+4L\{v(t)\}=7L\{e^{-t}\}$$

$$\left[s^2V(s)-sv(0)-v'(0)\right]+4\left[sV(s)-v(0)\right]+4V(s)=7\left(\frac{1}{s+1}\right)$$

Substitute 2 for $v(0)$ and 2 for $v'(0)$ in the equation.

$$\left[s^2V(s)-s(2)-2\right]+4\left[sV(s)-2\right]+4V(s)=\frac{7}{s+1}$$

$$\left(s^2+4s+4\right)V(s)-2s-10=\frac{7}{s+1}$$

$$\left(s^2+4s+4\right)V(s)=\frac{7}{s+1}+(2s+10)$$

$$\left(s^2+4s+4\right)V(s)=\frac{(2s+10)(s+1)+7}{s+1}$$

Step 2 of 6

Further simplification as follows,

$$\left(s^2+4s+4\right)V(s)=\frac{2s^2+12s+17}{s+1}$$

$$V(s)=\frac{2s^2+12s+17}{(s+1)\left(s^2+4s+4\right)}$$

$$V(s)=\frac{2s^2+12s+17}{(s+1)(s+2)^2}$$

Apply partial fractions to simplify the equation $V(s)$.

$$\frac{2s^2+12s+17}{(s+1)(s+2)^2}=\frac{A}{(s+1)}+\frac{B}{(s+2)^2}+\frac{C}{(s+2)} \quad \ldots\ldots (1)$$

Step 3 of 6

Calculate the constant A using residue theorem.

$$A=\left[(s+1)V(s)\right]\Big|_{s=-1}$$
$$=\left[(s+1)\left(\frac{2s^2+12s+17}{(s+1)(s+2)^2}\right)\right]\Big|_{s=-1}$$
$$=\left[\frac{2s^2+12s+17}{(s+2)^2}\right]\Big|_{s=-1}$$
$$=\frac{2(-1)^2+12(-1)+17}{(-1+2)^2}$$
$$=\frac{2-12+17}{1}$$
$$=7$$

Step 4 of 6

Calculate the constant B using residue theorem.

$$B=\left[(s+2)^2V(s)\right]\Big|_{s=-2}$$
$$=\left[(s+2)^2\frac{2s^2+12s+17}{(s+1)(s+2)^2}\right]\Big|_{s=-2}$$
$$=\left[\frac{2s^2+12s+17}{(s+1)}\right]\Big|_{s=-2}$$
$$=\frac{2(-2)^2+12(-2)+17}{(-2+1)}$$
$$=\frac{8-24+17}{-1}$$
$$=-1$$

Step 5 of 6

Calculate the constant C using residue theorem.

$$C=\left[\frac{d}{ds}\left[(s+2)^2V(s)\right]\right]\Big|_{s=-2}$$
$$=\left[\frac{d}{ds}\left(\frac{2s^2+12s+17}{s+1}\right)\right]\Big|_{s=-2}$$
$$=\left[\frac{(s+1)(4s+12)-(2s^2+12s+17)}{(s+1)^2}\right]\Big|_{s=-2}$$
$$=\left[\frac{2s^2+4s-5}{(s+1)^2}\right]\Big|_{s=-2}$$
$$=\frac{2(-2)^2+4(-2)-5}{(-2+1)^2}$$
$$=-5$$

Step 6 of 6

Recall equation (1).

$$\frac{2s^2+12s+17}{(s+1)(s+2)^2}=\frac{A}{(s+1)}+\frac{B}{(s+2)^2}+\frac{C}{(s+2)}$$

$$V(s)=\frac{A}{(s+1)}+\frac{B}{(s+2)^2}+\frac{C}{(s+2)}$$

Substitute 7 for A, -1 for B and -5 for C in the equation.

$$V(s)=\frac{7}{(s+1)}+\frac{-1}{(s+2)^2}+\frac{-5}{(s+2)}$$

$$V(s)=\frac{7}{(s+1)}-\frac{1}{(s+2)^2}-\frac{5}{(s+2)}$$

Apply inverse transform on both sides,

$$L^{-1}\{v(t)\}=L^{-1}\left\{\frac{7}{s+1}\right\}-L^{-1}\left\{\frac{1}{(s+2)^2}\right\}-L^{-1}\left\{\frac{5}{s+2}\right\}$$

$$L^{-1}\{v(t)\}=7L^{-1}\left\{\frac{1}{s+1}\right\}-L^{-1}\left\{\frac{1}{(s+2)^2}\right\}-5L^{-1}\left\{\frac{1}{s+2}\right\}$$

It is known that,

$$L^{-1}\left\{\frac{1}{s+a}\right\}=e^{-at}$$

$$L^{-1}\left\{\frac{1}{(s+2)^2}\right\}=te^{-2t}$$

Therefore,

$$\begin{aligned} v(t)&=7e^{-t}u(t)-te^{-2t}u(t)-5e^{-2t}u(t) \\ &=\left(7e^{-t}-te^{-2t}-5e^{-2t}\right)u(t) \end{aligned}$$

Thus, the expression for $v(t)$ is, $\boxed{\left(7e^{-t}-5e^{-2t}-te^{-2t}\right)u(t)}$.